

DANIEL NGUYEN

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EDUCATION

University of Ottawa

Ottawa, ON

BASc in Computer Engineering (Co-op), Third Year — CGPA 6.95

Sept. 2024 – Apr. 2029 (expected)

- **Coursework:** Data Structures & Algorithms, Software Engineering, Computer Architecture I & II, Digital Systems II, Product Development

EXPERIENCE

Intern Solutions Engineer (Co-op)

May 2026 – Aug. 2026

Teldio Corporation — Integration Engineering for an Event-Driven Middleware Platform

Ottawa, ON

- Built **Project Echo**, an internal tool that captures live protocol traffic from integration modules and replays it as automated test input, removing a physical-hardware bottleneck from the QA cycle; still in use after the term
- Extended Echo's capture and decoding to MOTOTRBO radio and Modbus, wired it into the automated test suite, and integrated results with TestRail and Jira
- Stood up four C-CURE 9000 VMs spanning vendor versions and integrated against all of them over both the REST API and the CrossFire .NET SDK (bridged into Go modules, byte-identical events); handed the PoC to R&D
- Shipped and demoed the Avigilon Unity Access module with an MQTT/JSON/TLS event relay (10 actions, 14 event types, 4 transports, 121 tests); found an MQTT 3.1.1 conformance defect reported upstream to Motorola
- Integrated proprietary vendor systems, the DSC PowerSeries Neo (ITv2) and Bosch B4512 (Mode 2) intrusion panels, from protocol research to proof of concept; Bosch delivered ahead of schedule with a 93-item test checklist and a PoC linking alarm events to a camera system

Computer Technician (Co-op)

Mar. 2023 – June 2023

XPC

Toronto, ON

- Repaired 50+ desktops, laptops, and servers, resolving OS and hardware faults with a ~90% recovery rate
- Built and optimized 20+ custom PCs; delivered SSD/RAM upgrade recommendations that improved reliability

PROJECTS

MacTrack | Kotlin, Jetpack Compose, Room, CameraX, ML Kit

June 2026 – Sept. 2026

- Offline-first Android calorie and macro tracker; Room database evolved to schema v8 through eight non-destructive migrations with zero data loss
- Three-source food search (bundled Canadian Nutrient File, user foods, Open Food Facts) and fully offline barcode scanning with CameraX and ML Kit
- Opt-in BYO-key AI assistant with the key encrypted in the Android Keystore; unit-tested TDEE engine, GitHub Actions CI on every push

Homelab | Docker Compose, Caddy, Tailscale, systemd, Linux

July 2026 – Present

- 11 containers across 7 Compose stacks on an immutable Fedora host: photo library, GPU LLM inference, browser game streaming to a TV
- Every service behind a custom Caddy build with a wildcard Let's Encrypt certificate (DNS-01), loopback-only ports, zero inbound exposure; Tailscale /32 route for identical remote access
- Fixed a container running inference on CPU for six weeks (~10x on GPU) and a clean-exit service that never bound its port (socket-checking systemd timer); both in the incident log

SKILLS & ACTIVITIES

Programming: Python, Java, Kotlin, Go, PowerShell, Bash; learning C, C++ , Rust

Frameworks & Protocols: Android, Firebase, REST APIs, Node-RED, MOTOTRBO

Tools: Git, GitHub Actions, Docker Compose, Android Studio, Jira, TestRail, Proxmox/VMware, OnShape, Fusion 360

Infrastructure: Linux, Caddy, Let's Encrypt/TLS, DNS, Tailscale/WireGuard, SSH, Windows

Hardware: Soldering, circuit design, PCB assembly, microcontrollers (Arduino/ESP), motor control, sensor integration

Activities: Intramural volleyball (uOttawa)

Languages: English, Vietnamese

Interests: Robotics, embedded systems, computer hardware, gym, volleyball, nutrition